The empirical recurrence for OEIS A303411, recovered from its tail

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Abstract

OEIS A303411 records "Empirical recurrence of order 71" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 73$. Its last coefficient is $c_{71} = 64 \neq 0$, so no further tail satisfies anything shorter; any order-71 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A303411 is "Number of 4Xn 0..1 arrays with every element equal to 1, 2, 4, 5 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$0, 23, 119, 168, 547, 2079, 6322, 17903, \ \ldots$$

The entry states:

> Empirical recurrence of order 71 (see link above)

As of the "Last modified" line on the live entry (Apr 23 2018 10:42:41, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^{\top} M^{j}$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 71: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 71.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 71 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{71} c_i\, a(n-i),$$

c_1	9	c_{19}	133490	c_{37}	23791825	c_{55}	−12307824
c_2	−25	c_{20}	127201	c_{38}	34465219	c_{56}	−9080478
c_3	11	c_{21}	364272	c_{39}	−30622691	c_{57}	−2794000
c_4	47	c_{22}	588740	c_{40}	36901004	c_{58}	−3466285
c_5	−114	c_{23}	−250856	c_{41}	−8589088	c_{59}	−582283
c_6	220	c_{24}	−1222710	c_{42}	−75713977	c_{60}	1737263
c_7	419	c_{25}	−1919523	c_{43}	11139195	c_{61}	1271489
c_8	−1631	c_{26}	−1537150	c_{44}	−17660428	c_{62}	875704
c_9	−1095	c_{27}	38470	c_{45}	120046108	c_{63}	400396
c_{10}	4778	c_{28}	2551177	c_{46}	49597818	c_{64}	54646
c_{11}	−2626	c_{29}	4907493	c_{47}	−62863706	c_{65}	−17236
c_{12}	−709	c_{30}	6706518	c_{48}	−41893264	c_{66}	−28036
c_{13}	9509	c_{31}	614189	c_{49}	−132601857	c_{67}	−13040
c_{14}	8693	c_{32}	4255166	c_{50}	1927144	c_{68}	−1264
c_{15}	−17897	c_{33}	−15481338	c_{51}	138181490	c_{69}	752
c_{16}	−25954	c_{34}	−16181652	c_{52}	42268502	c_{70}	448
c_{17}	−50864	c_{35}	−9511593	c_{53}	−24135486	c_{71}	64
c_{18}	−79609	c_{36}	−3191807	c_{54}	−12683612		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 73$, and it is the recurrence OEIS A303411 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{71} - c_1 x^{71-1} - \cdots - c_{71}$. Its constant term is $-c_{71} = -64$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 71 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 71, so $\deg q = 71$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 71, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 71, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 64.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-71 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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